# WEEK 8 PROGRESS REPORT

## Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors

ECE501: Digital Image Processing | GROUP 19 | November 9, 2026

### Team Members


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### Executive Summary


This week extended the fabric-defect pipeline by evaluating Gabor descriptors with square and thin rectangular windows, adding a GLCM anomaly-score branch, comparing AND/OR/weighted Gabor–GLCM fusion, and testing the trained pipeline on a newly uploaded fabric strip with its ground-truth mask. Among the tested Gabor windows, 8×32 produced the highest reported Dice (0.7189) and IoU (0.5612), narrowly exceeding the 16×16 result (Dice 0.7087).

The fusion experiment did not improve the overall balance of the Gabor-only model. Weighted fusion (Gabor weight 0.50) slightly increased precision and reduced the false-alarm rate, but recall and Dice dropped substantially. The separate new-fabric test produced precision 0.0012, false-alarm rate 0.9989 and Pixel AUC 0.5104, indicating that the current threshold/fusion setup does not transfer reliably to that sample. These results are treated as diagnostic findings, not as evidence that the final detector is ready for deployment.


### Objectives and Scope

- Evaluate thin Gabor windows (8×32 and 32×8) alongside the square windows (8×8, 16×16 and 32×32).
- Run a GLCM anomaly branch and compare it with Gabor using mask-based metrics.
- Compare Gabor-only, GLCM-only, AND, OR and weighted Gabor–GLCM fusion.
- Test the trained model on a new fabric strip and its corresponding ground-truth mask.
- Analyse false alarms, missed defects, threshold behaviour and localization quality.

# 1. Experimental Setup and Methodology

## 1.1 Dataset handling carried forward

The Week 7 protocol used Fabric Type 03 strips of size 256×4096 and divided each strip into 256×256 tiles without resizing. This avoided the horizontal distortion caused by resizing an entire strip to 256×256. The Week 7 dataset contained 20 defect-free strips and 30 defective strips with masks (31 defective strips were available, with one lacking a mask). Normal strips were split by source strip into reference, calibration and normal-test groups to reduce tile-level data leakage; defective strips were divided by source strip into validation and test groups.

Week 7 also identified tiles 0–3 as pure-white edge tiles and excluded them. Tile 4 was partly white and contained 5.1% of the defect pixels, so removing it improved the reported operating point but also reduced defect coverage. A paired tile-4 experiment is still required to isolate the effect of calibration changes from the effect of changing the test set.

## 1.2 Gabor texture descriptor

The Week 7 Gabor setup used 12 filters (three wavelengths × four orientations). For each local window, the mean and standard deviation of the Gabor magnitude responses were used as texture features, and a Mahalanobis distance from the normal reference was used as an anomaly score. Week 8 extended the window study to include rectangular windows, allowing the descriptor to capture texture variation differently along the two fabric directions.

## 1.3 GLCM anomaly branch and fusion

A separate GLCM anomaly-score map was generated and evaluated alongside the Gabor map. The two score maps were then combined using AND, OR and weighted fusion. The reported weighted experiment used a Gabor weight of 0.50.

## 1.4 Evaluation metrics

Precision measures how many predicted defect pixels are true defects; recall measures how many ground-truth defect pixels are detected. IoU and Dice measure overlap between the predicted and ground-truth masks. The false-alarm rate is reported as FP/(FP+TN). Pixel AUC measures how well anomaly scores rank defect pixels above normal pixels and is not tied to one binary threshold. Therefore, a high Pixel AUC can coexist with poor Dice if the selected threshold produces an unsuitable binary mask.

# 2. Gabor Window-Size Experiments

Five window shapes were evaluated. The table below transcribes the values displayed in the Week 8 notebook output. 'Validation Dice' is kept separate from the subsequent thresholded metrics, as labelled in the output.

| Window | Threshold | Validation Dice | Precision | Recall | IoU | Dice | Pixel AUC | Normal pixels flagged |
|---|---|---|---|---|---|---|---|---|
| 8×8 | 0.9814 | 0.1246 | 0.4564 | 0.5624 | 0.3368 | 0.5039 | 0.9286 | 2.3538% |
| 16×16 | 0.9995 | 0.1106 | 0.8510 | 0.6071 | 0.5488 | 0.7087 | 0.9613 | 0.3735% |
| 32×32 | 1.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.9793 | 0.0000% |
| 8×32 | 0.9992 | 0.1865 | 0.8842 | 0.6057 | 0.5612 | 0.7189 | 0.9484 | 0.2787% |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 32×8 | 1.0000 | 0.0737 | 0.9194 | 0.0655 | 0.0651 | 0.1222 | 0.9669 | 0.0202% |

## 2.1 Interpretation

8×32 achieved the best reported thresholded localization: Dice 0.7189 and IoU 0.5612. Its precision was 0.8842 and recall was 0.6057, with 0.2787% of normal pixels flagged.

The 8×32 result slightly exceeded 16×16 (Dice 0.7087, IoU 0.5488). This is a promising improvement, but the difference is small and should be checked across source-strip folds before concluding that 8×32 is consistently better.

32×8 produced the highest precision (0.9194) and very few normal pixels flagged (0.0202%), but recall was only 0.0655. It detects very few of the actual defect pixels and is therefore too conservative as a stand-alone setting.

32×32 had the highest Pixel AUC (0.9793), but its threshold was 1.0000 and all thresholded metrics were zero. This indicates a threshold/operating-point problem: strong ranking does not guarantee a useful binary defect mask.

8×8 was less precise and flagged more normal pixels (2.3538%) than the larger windows.

```
Training Gabor window: (8, 8)

Training Gabor window: (16, 16)

Training Gabor window: (32, 32)

Training Gabor window: (8, 32)

Training Gabor window: (32, 8)
```

| | Window | Threshold | Validation Dice | Precision | Recall | IoU | Dice | Pixel AUC | Normal pixels flagged % |
|---|---|---|---|---|---|---|---|---|---|
| 0 | 8x8 | 0.9814 | 0.1246 | 0.4564 | 0.5624 | 0.3368 | 0.5039 | 0.9286 | 2.3538 |
| 1 | 16x16 | 0.9995 | 0.1106 | 0.8510 | 0.6071 | 0.5488 | 0.7087 | 0.9613 | 0.3735 |
| 2 | 32x32 | 1.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.9793 | 0.0000 |
| 3 | 8x32 | 0.9992 | 0.1865 | 0.8842 | 0.6057 | 0.5612 | 0.7189 | 0.9484 | 0.2787 |
| 4 | 32x8 | 1.0000 | 0.0737 | 0.9194 | 0.0655 | 0.0651 | 0.1222 | 0.9669 | 0.0202 |

*Figure 1. Notebook output for the Gabor window-size comparison, including square and thin rectangular windows.*

## 3. GLCM and Gabor–GLCM Fusion Results

The GLCM branch was evaluated on its own and in combination with Gabor. The weighted fusion row used a Gabor weight of 0.50. The metrics below are taken directly from the supplied fusion comparison output.

| Model | Precision | Recall | IoU | Dice | False-alarm rate | Pixel AUC | Image AUC | Threshold |
|---|---|---|---|---|---|---|---|---|
| Gabor only | 0.8510 | 0.6071 | 0.5488 | 0.7087 | 0.0037 | 0.9613 | 0.9107 | 0.9995 |
| GLCM only | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.8637 | 0.7566 | 1.0000 |
| Gabor + GLCM (AND) | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.9250 | 0.8971 | — |
| Gabor + GLCM (OR) | 0.8510 | 0.6071 | 0.5488 | 0.7087 | 0.0037 | 0.9466 | 0.8545 | — |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Gabor + GLCM (weighted, 0.50) | 0.8658 | 0.2136 | 0.2067 | 0.3426 | 0.0012 | 0.9415 | 0.8933 | 0.9997 |

## 3.1 Confusion-count comparison

| Model | TP | FP | FN | TN |
|---|---|---|---|---|
| Gabor only | 108,061 | 18,915 | 69,922 | 5,045,982 |
| GLCM only / AND | 0 | 0 | 177,983 | 5,064,897 |
| Gabor + GLCM OR | 108,061 | 18,915 | 69,922 | 5,045,982 |
| Weighted fusion (0.50) | 38,011 | 5,893 | 139,972 | 5,059,004 |

## 3.2 Interpretation

Gabor-only was the strongest overall thresholded model in this comparison, with Dice 0.7087 and recall 0.6071.

Weighted fusion raised precision from 0.8510 to 0.8658 and reduced the false-alarm rate from 0.0037 to 0.0012. However, recall fell from 0.6071 to 0.2136 and Dice fell from 0.7087 to 0.3426. The model became more conservative but missed substantially more defect pixels.

GLCM-only produced no positive predictions at the reported threshold of 1.0000, even though its Pixel AUC was 0.8637. This suggests that the GLCM score contains some ranking information, but the current threshold is not producing a useful binary mask.

AND fusion also produced no positive predictions, consistent with the GLCM branch suppressing detections when both binary masks are required to agree.

OR fusion had the same thresholded Precision, Recall, IoU and Dice as Gabor-only. Under the tested settings, adding the GLCM binary mask did not increase the number of useful detections.

Only one weighted-fusion weight (0.50) is shown. It is an initial test, not a completed weight optimization. A systematic weight sweep must be performed on validation strips.

| | Precision | Recall | IoU | Dice | False alarm rate | TP | FP | FN | TN | Model | Threshold | Validation Dice | Pixel AUC | Image AUC | Gabor threshold | GLCM threshold | Weight on Gabor |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0.8510 | 0.6071 | 0.5488 | 0.7087 | 0.0037 | 108061 | 18915 | 69922 | 5045982 | Gabor only | 0.9995 | 0.1106 | 0.9613 | 0.9107 | NaN | NaN | NaN |
| 1 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0 | 0 | 177983 | 5064897 | GLCM only | 1.0000 | 0.0000 | 0.8637 | 0.7566 | NaN | NaN | NaN |
| 2 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0 | 0 | 177983 | 5064897 | Gabor + GLCM AND | NaN | 0.0000 | 0.9250 | 0.8971 | 0.9995 | 1.0 | NaN |
| 3 | 0.8510 | 0.6071 | 0.5488 | 0.7087 | 0.0037 | 108061 | 18915 | 69922 | 5045982 | Gabor + GLCM OR | NaN | 0.1106 | 0.9466 | 0.8545 | 0.9995 | 1.0 | NaN |
| 4 | 0.8658 | 0.2136 | 0.2067 | 0.3426 | 0.0012 | 38011 | 5893 | 139972 | 5059004 | Gabor + GLCM weighted | 0.9997 | 0.1252 | 0.9415 | 0.8933 | NaN | NaN | 0.5 |

*Figure 2. Fusion comparison output showing Gabor-only, GLCM-only, AND, OR and weighted fusion.*

**Fusion conclusion:** The current evidence does not support replacing Gabor-only with weighted Gabor–GLCM fusion. The next experiment should tune the fusion weight and each branch's threshold on validation data, then evaluate the frozen configuration on the held-out test strips.

## 4. Testing on a Newly Uploaded Fabric Strip

After model training, a new fabric strip and its corresponding ground-truth mask were uploaded for testing. The displayed tile visualizations use a Gabor weight of 0.50 and a threshold of 0.9997. The metric table reports the following results for the new sample:

| Metric | New-sample result |
|---|---|
| Precision | 0.0012 |
| Recall | 1.0000 |
| IoU | 0.0012 |
| Dice | 0.0024 |
| False-alarm rate | 0.9989 |
| TP | 861 |
| FP | 719,267 |
| FN | 0 |
| TN | 768 |
| Pixel AUC | 0.5104 |

### 4.1 Interpretation of the new-sample result

The recall of 1.0000 means that the reported comparison found all ground-truth defect pixels, but this came with an extremely large number of false positives. Precision was only 0.0012 and the false-alarm rate was 0.9989 (approximately 99.89% of normal pixels were flagged). Dice was 0.0024 and Pixel AUC was 0.5104, which is close to chance-level ranking for this sample.

The confusion counts contain 720,896 pixels in total, equivalent to eleven 256×256 tiles. The result therefore indicates that the current operating threshold is not suitable for this new strip. It should not be presented as successful defect localization, even though recall is 1.0: the detector is flagging almost everything as defective.

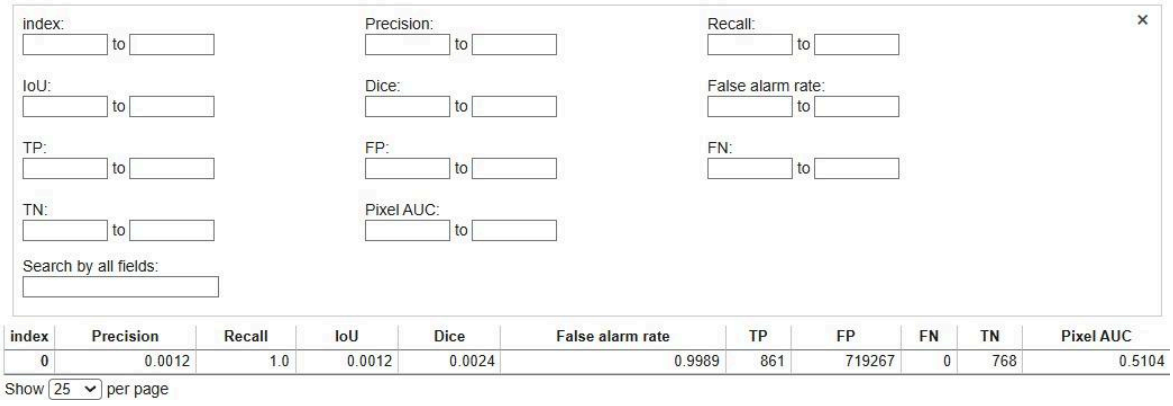



*Figure 3. Quantitative evaluation table for the newly uploaded fabric sample.*

## 4.2 Visual observations

The tile visualizations show that GLCM anomaly maps contain localized, block-like high-response regions, while the Gabor maps appear very dark or near-zero in several displayed tiles. The fused score often resembles the GLCM response. In the separate example with a thin defect marked in the ground-truth mask, the weighted prediction does not visibly reproduce the full thin defect. This agrees with the poor Dice result from the new-sample evaluation.

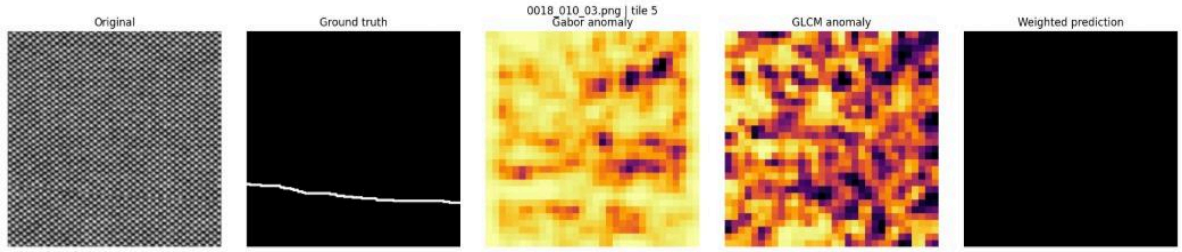



*Figure 4. Example tile: original fabric, ground-truth mask, Gabor anomaly map, GLCM anomaly map and weighted prediction.*

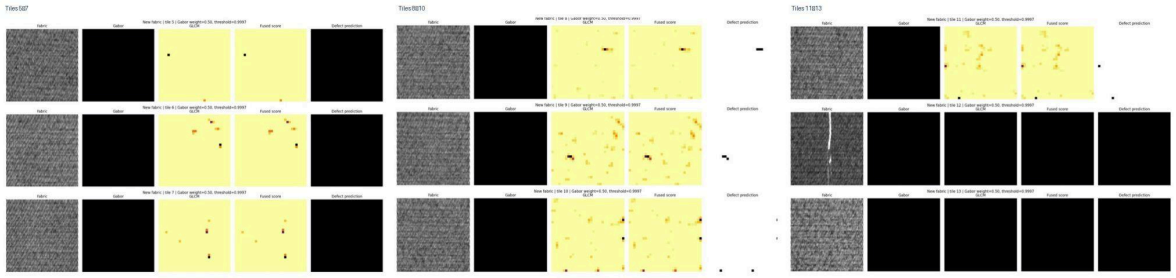


*Figure 5. Additional qualitative outputs for the newly tested fabric strip (tiles 5–13).*

## 4.3 Checks required before accepting the new-sample result

- Confirm that the uploaded mask belongs to the exact fabric image and has the same dimensions, orientation and tile alignment.
- Confirm the mask convention (defect = 1/white and normal = 0/black) is used consistently in both metric calculation and visualization.
- Check whether the threshold operation is score > threshold or score ≥ threshold, especially for models whose selected threshold is exactly 1.0000.
- Inspect confusion counts separately for each tile. A strip-level aggregate can hide tiles with no detections and other tiles with excessive detections.
- Verify that score normalization is consistent between reference, calibration, validation, held-out test and new-fabric inference.
- Choose thresholds and fusion weights using validation strips only; do not tune them using the held-out test mask.

# 5. What Week 8 Shows

- Thin rectangular windows are worth keeping in the study. The 8×32 configuration produced the best reported Dice (0.7189) and IoU (0.5612) in the supplied Gabor window table.
- The 32×8 window is highly conservative: precision is high, but recall is too low for reliable defect localization.

- The GLCM branch produces anomaly-score variation, but its current binary threshold is not effective. GLCM-only and AND fusion produced no positive predictions in the reported comparison.
- The fixed 0.50 weighted fusion reduces false alarms but sacrifices too much recall and Dice compared with Gabor-only.
- The new-fabric test reveals a serious transfer/calibration problem: false-alarm rate 0.9989 and Pixel AUC 0.5104. The detector is not yet reliable on this unseen sample.
- The visual maps suggest that the current pipeline does not consistently trace thin or elongated defects, so localization quality must be checked in addition to image-level detection.

## 7. Plan for Week 9

- **Debug and validate the new-sample evaluation.** Check image-mask pairing, mask polarity, image orientation, tile indexing, dimensions, threshold direction and per-tile confusion counts. Re-run the sample only after these checks.
- **Tune thresholds and fusion weights on validation strips.** Run a controlled grid for GLCM thresholds and Gabor weights (for example 0.00, 0.25, 0.50, 0.75 and 1.00). Select the operating point using validation Dice or a stated precision/recall constraint, then freeze it before testing.
- **Complete the paired tile-4 experiment.** Use the same exact held-out test tiles in both conditions. Change only whether tile 4 is included in calibration, and report the separate coverage cost of excluding tile 4.
- **Run cross-validation by source strip.** Keep all tiles from one source strip in the same fold to avoid leakage. Report mean ± standard deviation for Dice, IoU, Pixel AUC, precision, recall and false-alarm rate.
- **Report per-defect-type recall and cTPR.** Break down performance by defect category and clearly define the cTPR calculation and operating point.
- **Reproduce the published-AITEX comparison.** Use the same dataset subset, preprocessing, split and metric definitions as the referenced published method wherever possible. State any differences.
- **Train and evaluate the two-class baseline.** Use the same source-strip split and masks so that the supervised two-class result can be compared fairly with the anomaly-based models.
- **Document the GLCM implementation completely.** Record gray-level quantization, offsets/directions, window size, features, normal reference construction and the exact role of Enhanced_KNN and Enhanced_Mahalanobis.

## 8. Conclusion

Week 8 extended the project beyond the independent Gabor baseline by testing thin rectangular windows, evaluating a GLCM anomaly branch and comparing several fusion rules. The 8×32 Gabor window gave the strongest reported Dice score in the window sweep, while the fixed weighted fusion did not improve overall localization compared with Gabor alone. The new-fabric test exposed a major false-alarm and transfer problem, so threshold calibration, mask alignment and per-tile diagnostics must be addressed before claiming robust generalization. Week 9 will focus on validating the evaluation pipeline, completing the pending controlled comparisons and reporting performance across source-strip folds.